

BANARAS HINDU UNIVERSITY

B.Sc. (Hons.) — Semester V Examination, 2013-14

Paper No. BSC-05A: Elements of Earth Science (Ancillary)

Time: 3 Hours

Maximum Marks: 70

Ref. No.: BS/Sem V/260

*Instructions: Attempt any **five** questions, selecting at least **two** questions each from Section A and Section B. Section C is compulsory. All questions carry equal marks (14 marks each).*

SECTION — A

Question 1. Describe the geological work of a glacier. [14]

Question 2. Write a detailed note on **any two** of the following: [2 × 7 = 14]

(a) Mountain building

(b) Plate Tectonics

(c) Rock cycle

Question 3. Discuss the significance of Earth System science with a focus on mitigating natural disasters. [14]

Question 4. What are igneous rocks? Write an essay on their classification giving suitable examples. [14]

SECTION — B

Question 5. Discuss the various causes for the generation of earthquakes giving suitable examples. [14]

Question 6. Write short notes on **any four** of the following: [4 × 3.5 = 14]

(a) Cirque

(b) Attrition

(c) Chemical weathering

(d) Alluvial fan

(e) Delta

(f) Present is Key to the past

Question 7. How do you distinguish between **any two** of the following mineral pairs based on their diagnostic physical properties? [2 × 7 = 14]

- (a) Muscovite and biotite
- (b) Calcite and quartz
- (c) Olivine and augite
- (d) Tourmaline and hornblende

SECTION — C (Compulsory)

Question 8. Explain the following terms in two or three sentences: [14 × 1 = 14]

- (a) Strike
- (b) Types of plate boundaries
- (c) Panthalassa
- (d) Schist
- (e) Sandstone
- (f) Paleontology
- (g) Lustre
- (h) Hardness in minerals
- (i) Asthenosphere
- (j) Syncline
- (k) Net slip
- (l) Exfoliation
- (m) Slate
- (n) Isostasy